NOTES ON AUTHENTICATION, AUTHORIZATION, AND ACCOUNTING (AAA)

- This is crucial security framework that controls and monitors user access to network resources.

How AAA mitigates vulnerabilities in a network

- **Authentication:** verifies the identity of user or device requesting access, typically, using credentials such as passwords, tokens, or biometrics
- **Authorization:** determines the permissions and what resources the authenticated user is allowed to access or actions they can perform.
- **Accounting:** Tracks and logs user activities for auditing, compliance, and resource usage monitoring

Importance of aaa in a network security against vulnerabilities

- It strengthens network security by ensuring that only legitimate users login or gain access to resources, enforcing access control policies and maintaining an audit trail of user actions for accountability and security investigations.
- It helps in compliance with security standards and detecting unauthorised activities

Protocols and implantation

- **Remote Authentication Dial-In User Service (RADIUS):** Provides centralized authentication, authorization, and accounting primarily for remote network access
- **Terminal Access Controller Access – Control System Plus (TACACS+)**: Offers similar functions with additional support for command – level authorization and device administration.

Differences between RADIUS and TACACS+

Features	RADIUS	TACACS+
Protocol	Uses UDP	Uses TCP
Transport characteristics	Connectionless faster but less reliable	Connection oriented, more reliable
Encryption	Encrypts only the password in the packet	Encrypts the entire packet payload
Authentication and authorization	Combined process	Separate processes for authentication, authorization and accounting
Accounting support	Strong accounting capabilities	Accounting separate
Typical use	Remote user network access authentication	Device administration (routers, switches)
Vendor support	Open standard widely supported	CISCO proprietary but widely used
Authorised control	Limited command – level authorization	Granular control level over commands and device management
Use case	Best for authentication remote users (Wi-Fi, VPN)	Best for managing device administration and command authorization

Security benefits of AAA

1. Centralizes user credentials and access control, reducing errors and vulnerabilities caused by inconsistent or outdated local configurations
2. Encrypts authentication data to protect credentials from interception during network communication
3. Enables detailed access control and monitoring, which deters unauthorised actions and supports compliance with regulations
4. Provides redundancy and failover for authentication services, ensuring continuous enforcement of access policies

Drawbacks of AAA

1. Complexity and management: AAA is considered to be complicated to implement, maintain, and manage especially in large and diverse network environments, requiring coordinated configuration across multiple devices and systems
2. Single point of failure: AAA relies on centralized servers meaning if the servers fail or become unreachable, users may lose access to network resources, potentially, halting operations until restored.
3. Cost considerations: AAA often involves additional hardware and software licenses and ongoing administrative overhead, making it costly for smaller organisations or limited resource environments
4. Security risks: if AAA servers or protocols are not properly secured, they could become targets for attackers such as interception of credentials or Denial of service (DDoS)on authentication servers.
5. Impact on user experience: authentication and authorization processes may introduce login delays or complexity, which can frustrate users and affect productivity

However, regardless of the above challenges, AAA is still considered a fundamental network security technology, and careful planning, redundancy, and proper configuration may help mitigate most vulnerability issues in organisations

Configuration commands (CISCO - IOS)

- *aaa new-model* //this enables AAA on the device or hardware to turn on AAA functionality globally
- *aaa authentication login [METHOD LIST] [method1] [method2]…* //configures login authentication methods such as using RADIUS, TACACS+, Local databases in case of databases and so on.
- *aaa authorization exec [METHOD-LIST]* //controls authorization for specific command levels in EXEC of configuration modes.
- *aaa accounting [network /exec /commands] [METHODS-LIST*] //enables accounting for network access, EXEC sessions, or commands executed.

Server and user configurations

- *radius server [NAME] and tacacs-server host [IP-ADDRESS]* //defines the RADIUS or TACACS+ server IP addresses and shared keys
- *aaa group server radius [GROUP-NAME]* //creates server groups referencing multiple AAA servers

- username [USER] secret [PASSWORD] //defines local users when fallback authentication is configured

Interface and line configuration

- *line vty 0 4* //enter the configuration mode for virtual terminal lines
- *login authentication [METHOD-LIST*] //applies the AAA authentication method list to the line for login

Task

Write AAA commands to configure AAA with RADIUS as the Primary authentication, authorization and accounting method, considering a local database as a fallback, applied to VTY line for remote access

Answer

aaa new-model
Radius server RADIUS1
Address ipv4 192.168.1.0 auth-po key mysecretkey
aaa authentication login default gro
aaa authorization exec default group
aaa accounting exec default start – st line vty 0 4
Login authentication default